

REVIEW

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TYPES OF FIELDS

SCALAR - MAGNITUDE OR DIRECTION

EG. TEMPERATURE, DENSITY, ELEVATION

VECTOR - MAGNITUDE & DIRECTION

EG. VELOCITY, GRAVITY, ELECTRIC, MAGNETIC

TENSOR - STRESS, ELECTRICAL / THERMAL CONDUCTIVITY, STRAIN

MIXING RELATIONSHIPS.

WHAT DO THEY MEAN?

HOW DO WE KNOW WHICH ONE TO USE?

HOW DO THEY RELATE TO EACH OTHER?

ARITHMETIC

$$\bar{P}_a = \frac{\sum_{i=1}^n P_i w_i}{\sum_{i=1}^n w_i}$$

LARGEST

GEOMETRIC

$$\bar{P}_g = \left(\prod_{i=1}^n P_i^{w_i} \right)^{\frac{1}{\sum_{i=1}^n w_i}}$$

IN-BETWEEN

HARMONIC

$$\bar{P}_h = \frac{\sum_{i=1}^n w_i}{\sum_{i=1}^n \frac{w_i}{P_i}}$$

SMALLEST

VOIGHT - REUSS - HILL

$$\bar{P} = \frac{1}{2} (\bar{P}_a + \bar{P}_h) \quad \text{CLOSER TO GEOMETRIC}$$

HASHIN - STRIKMAN (EXTREME BOUNDS)

NOT REALLY AN AVERAGE, BUT MEANT TO INDICATE
MAXIMUM / MINIMUM POSSIBLE EFFECTIVE BULK "AVERAGE"

HS⁺

ARITHMETIC

GEOMETRIC $\updownarrow$ VRH

HARMONIC

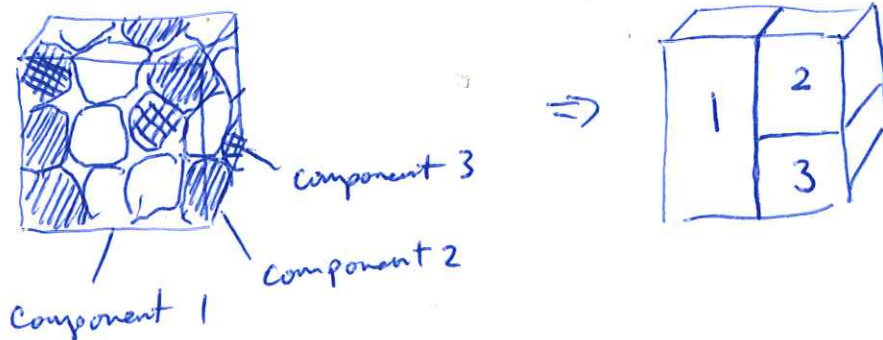
HS⁻

ARITHMETIC - WHEN TO USE?

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WHEN ALL COMPONENTS ARE EQUALLY IMPORTANT

EG. DENSITY

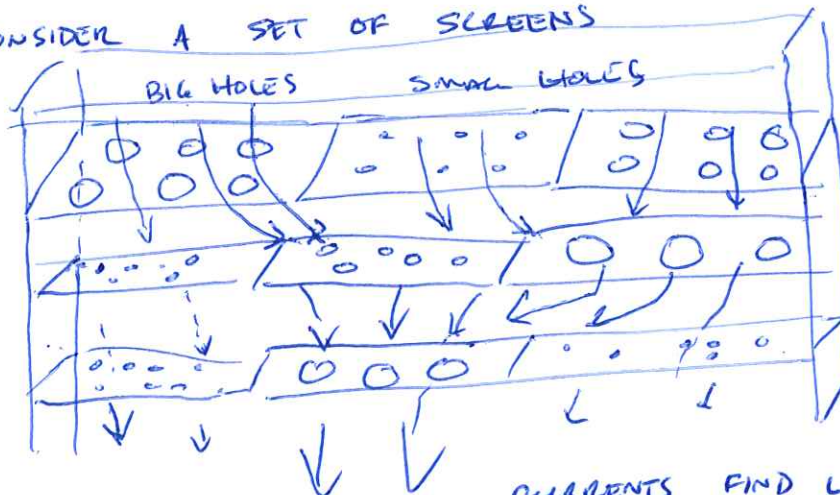


$$\begin{aligned}\bar{P} &= \frac{m_T}{V_T} = \frac{m_1 + m_2 + m_3}{V_1 + V_2 + V_3} \\ &= \frac{\rho_1 V_1 + \rho_2 V_2 + \rho_3 V_3}{V_1 + V_2 + V_3} \\ &= \frac{\sum_{i=1}^3 \rho_i V_i}{\sum_{i=1}^3 V_i}\end{aligned}$$

GEO METRIC - TYPICALLY WHEN THINGS ARE WELL MIXED

~~AND~~ BUT BIG VALUES MATTER MORE,

CONSIDER A SET OF SCREENS



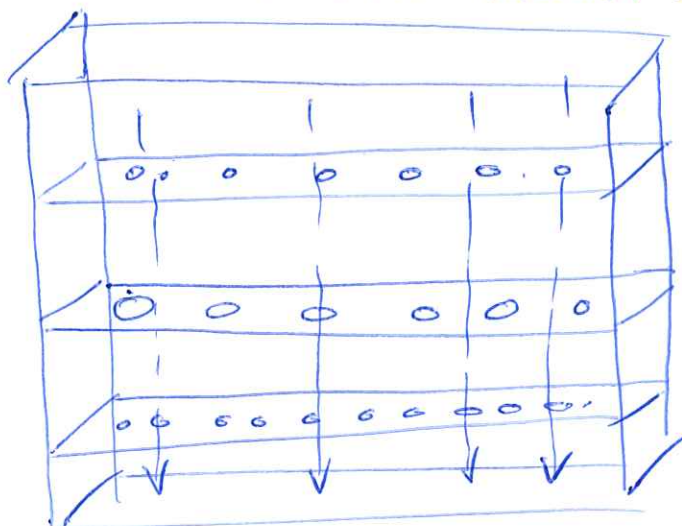
WATER
FLOWS
FAST THROUGH

BIG HOLES, SLOW
THROUGH SMALL
ONES.

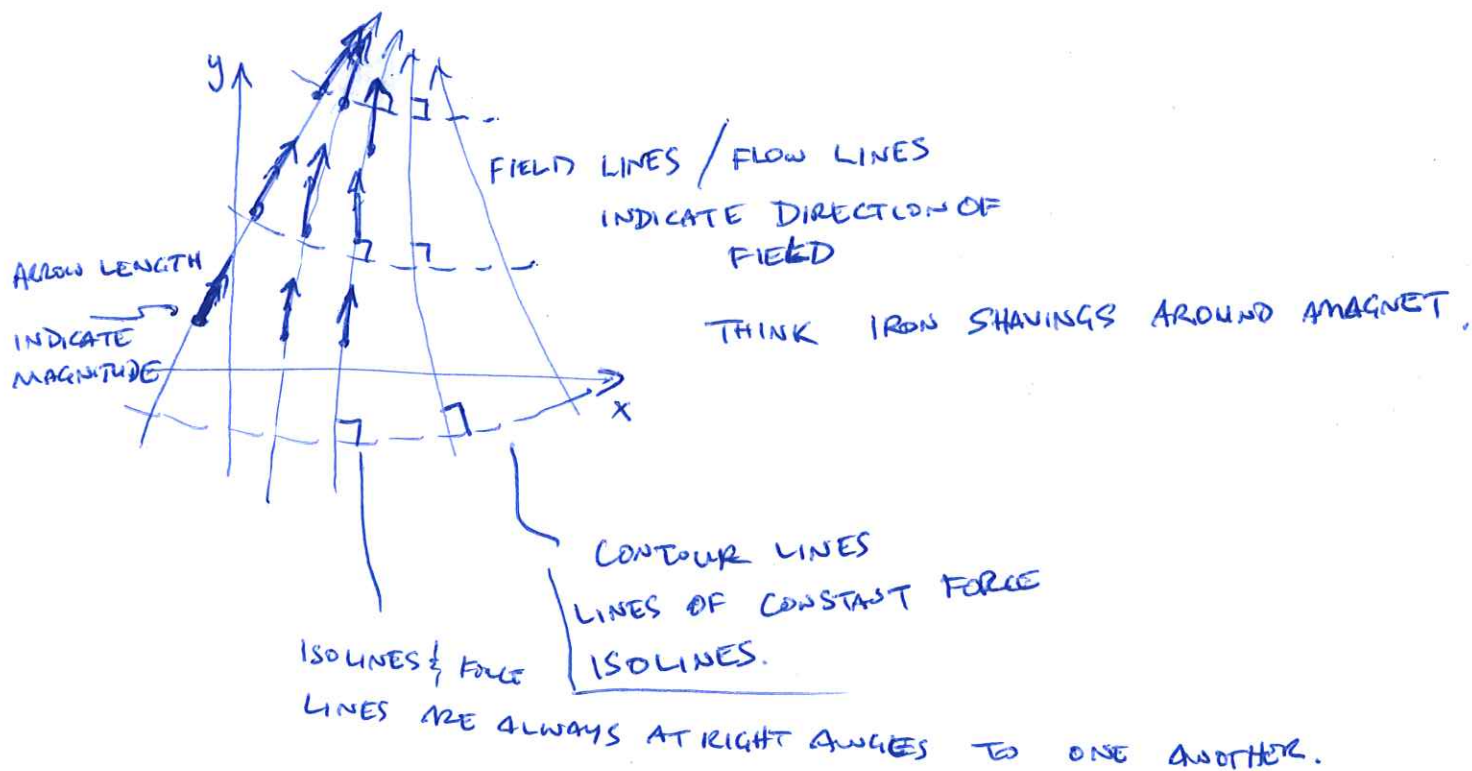
~~CONDUCTIVITY FINDS~~
CURRENTS FIND LEAST RESISTIVE (NOT SHORTEST)
PATH

HAZMONIC - WHEN FAST IS NOT AN OPTION
MUST TRAVEL THROUGH RESISTIVE AREAS...

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GRAPHICALLY REPRESENTING FIELDS



RELATIONSHIP BETWEEN WORK/ENERGY/POTENTIAL AND FORCE.
work done on charge or distance

$$F \Delta x = \Delta E = W \leftarrow \text{work or change in energy.}$$

$$F = \frac{\Delta E}{\Delta x} = \frac{W}{\Delta x}$$

$$F = \frac{dW}{dx} \leftarrow \text{THIS IS WHERE WE ARE GOING.}$$

↑
derivative with respect to position

DEFINITION OF FORCE

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$$\vec{F} = m \vec{a}$$

force mass acceleration

$\vec{}$ = vector

$\hat{}$ = unit vector (length = 1 magnitude)

$$\vec{a} = \frac{d\vec{v}}{dt}$$

$$\vec{F} = m \frac{d\vec{v}}{dt}$$

multiply both sides by $\vec{v}$

when multiplying vectors must use dot product.

$$\vec{F} \cdot \vec{v} = m \frac{d\vec{v}}{dt} \cdot \vec{v}$$

$$\frac{d}{dt} (\vec{v} \cdot \vec{v})$$

$$\frac{d}{dt} (\vec{v} \cdot \vec{v}) = \frac{d\vec{v}}{dt} \cdot \vec{v} + \vec{v} \cdot \frac{d\vec{v}}{dt}$$

chain rule

$$\vec{v} \cdot \vec{v} = v^2$$

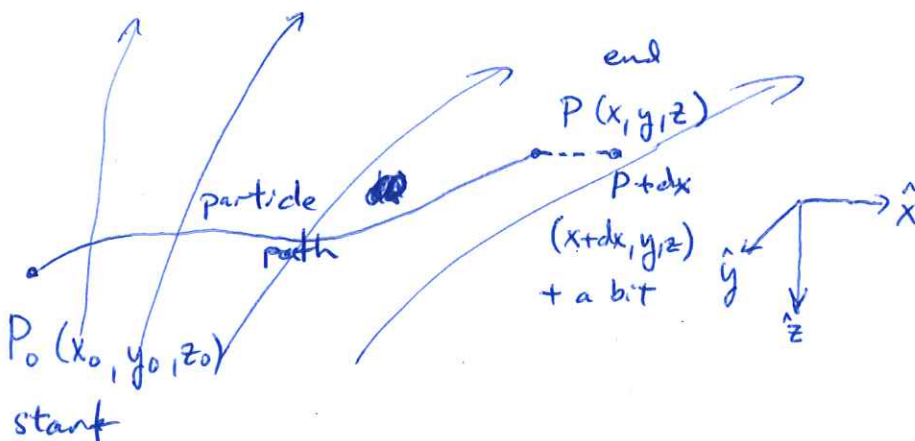
dot product scalar

$$\frac{d}{dt} (\vec{v} \cdot \vec{v}) = 2 \frac{d\vec{v}}{dt} \cdot \vec{v}$$

$$\frac{1}{2} \frac{dv^2}{dt} = \frac{d\vec{v}}{dt} \cdot \vec{v}$$

$$\vec{F} \cdot \vec{v} = \frac{1}{2} m \frac{dv^2}{dt} = \frac{dK}{dt}$$

the derivative kinetic energy



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$$\int_{t_0}^t \vec{F} \cdot \vec{v} dt' = K - K_0$$

↑ integral
 ↑ start time
 ← end time

$$\vec{v} dt' = d\vec{s}$$

↑ velocity
 ↑ some small time
 ← small distance

$$\int_{P_0}^P \vec{F} \cdot d\vec{s} = W - W_0 = W(P, P_0)$$

$$\underbrace{W(P, P_0)}_{\text{work from } P_0 \text{ to } P} + \underbrace{W(P+\Delta x, P)}_{\text{work from } P \text{ to } P+\Delta x} = \underbrace{W(P+\Delta x, P_0)}_{\text{work from } P_0 \text{ to } P+\Delta x}$$

path-independent

$$W(P+\Delta x, P) = W(P+\Delta x, P_0) - W(P, P_0)$$

$$\int_P^{P+\Delta x} F_x(x, y, z) dx = W(P+\Delta x, P_0) - W(P, P_0)$$

Continuity

A function is continuous if you can trace it without ever having to pick up a pen used to draw it.

There are no steps, breaks, holes, gaps, etc.

Differentiability

A continuous function that is well-behaved, with no changes in slope at a point.

A slope going into and out of the point is the same when approached from both sides.

if not, then we must choose a side

Principle of super-position

2 conditions to work

1. A function must be linear, i.e. 1 an add effect of field from one to the other

$$f(m_1 + m_2 + m_3 + \dots) = f(m_1) + f(m_2) + f(m_3) + \dots$$

2. A function must be homogeneous, i.e. making a mass larger or charge greater with ~~not change~~ change the function by a multiple of the change

$$f(am) = af(m)$$